Proprietary + Confidential



The extract, transform, and load data pipeline pattern

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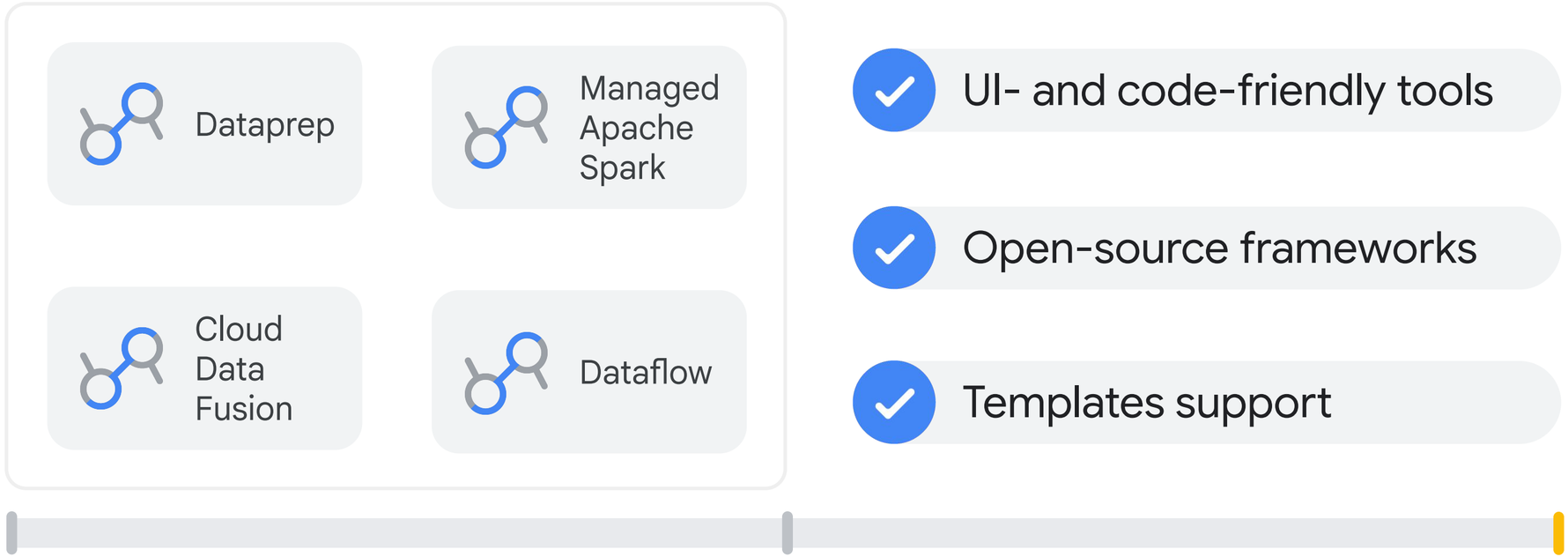
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01	Extract, transform, and load (ETL) architecture
02	Google Cloud GUI tools for ETL data pipelines
03	Batch data processing using Managed Service for Apache Spark
04	Streaming data processing options
05	Bigtable and data pipelines

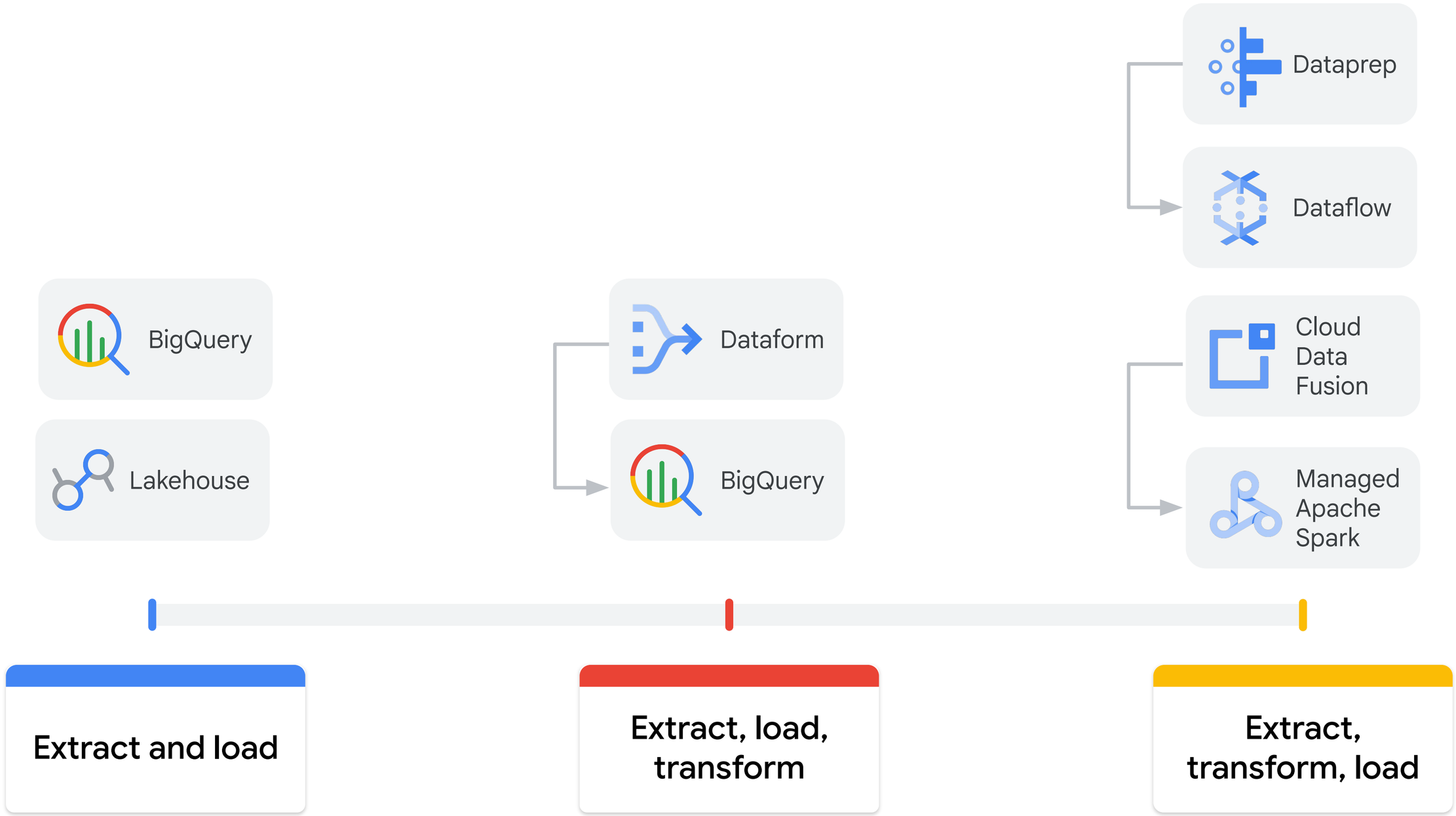


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Google Cloud provides multiple services for distributed data processing



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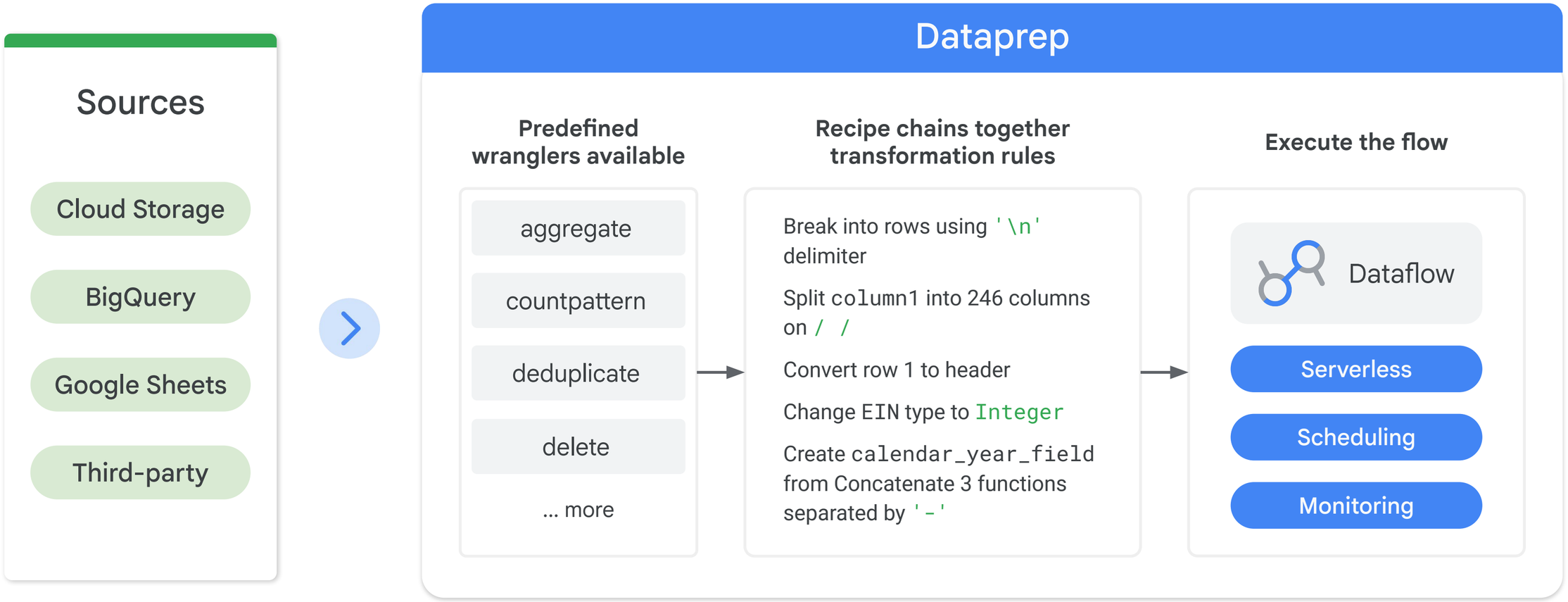
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Dataprep by Trifacta is a serverless, no-code solution to build data transformation flows



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Dataprep lets you preview your transformations and gives recommendations for your flow

Source should be dropped	Preview
Whse_Name	Whse_Name
	
2 Categories	1 Category
~~Whse 0~~	
~~Whse 0~~	
Acme Warehouse	Acme Warehouse
Acme Warehouse	Acme Warehouse
Acme Warehouse	Acme Warehouse
Acme Warehouse	Acme Warehouse
~~Whse 0~~	

Suggestions

Extract values matching

`Whse 0`

`Whse 0` starting after `{start}` ending before `{end}`

starting after `{start}` ending before `{end}`

Replace

`Whse 0` with "" in Whse_Name

Edit Add

Count values matching

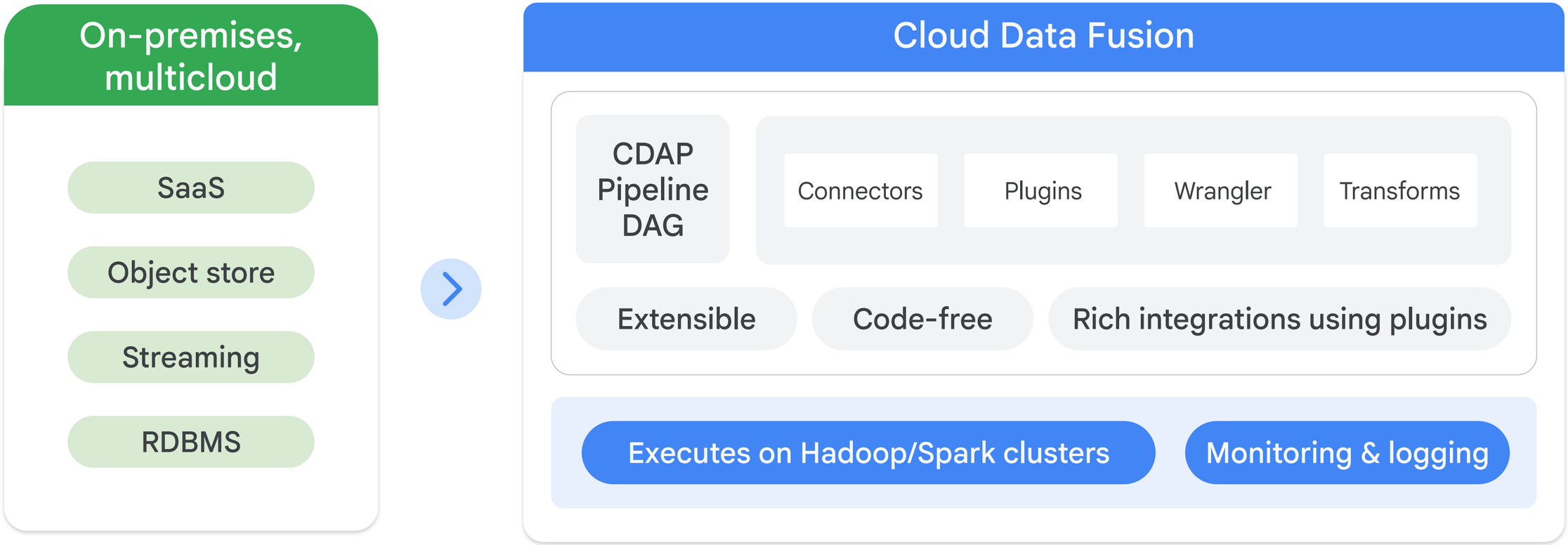
`Whse 0`

`Whse 0` starting after `{start}` ending before `{end}`

starting after `{start}` ending before `{end}`

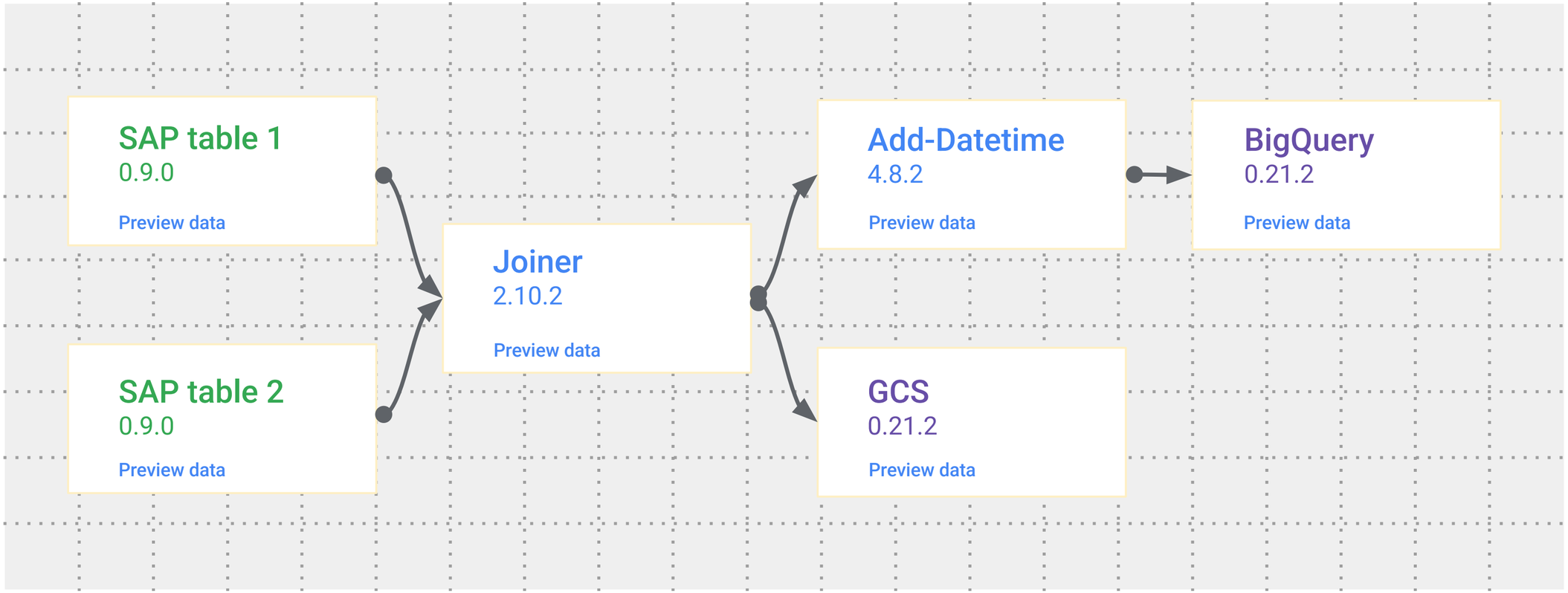
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Cloud Data Fusion is a GUI-based enterprise data integration service



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Create and deploy data integration pipelines in Cloud Data Fusion Studio



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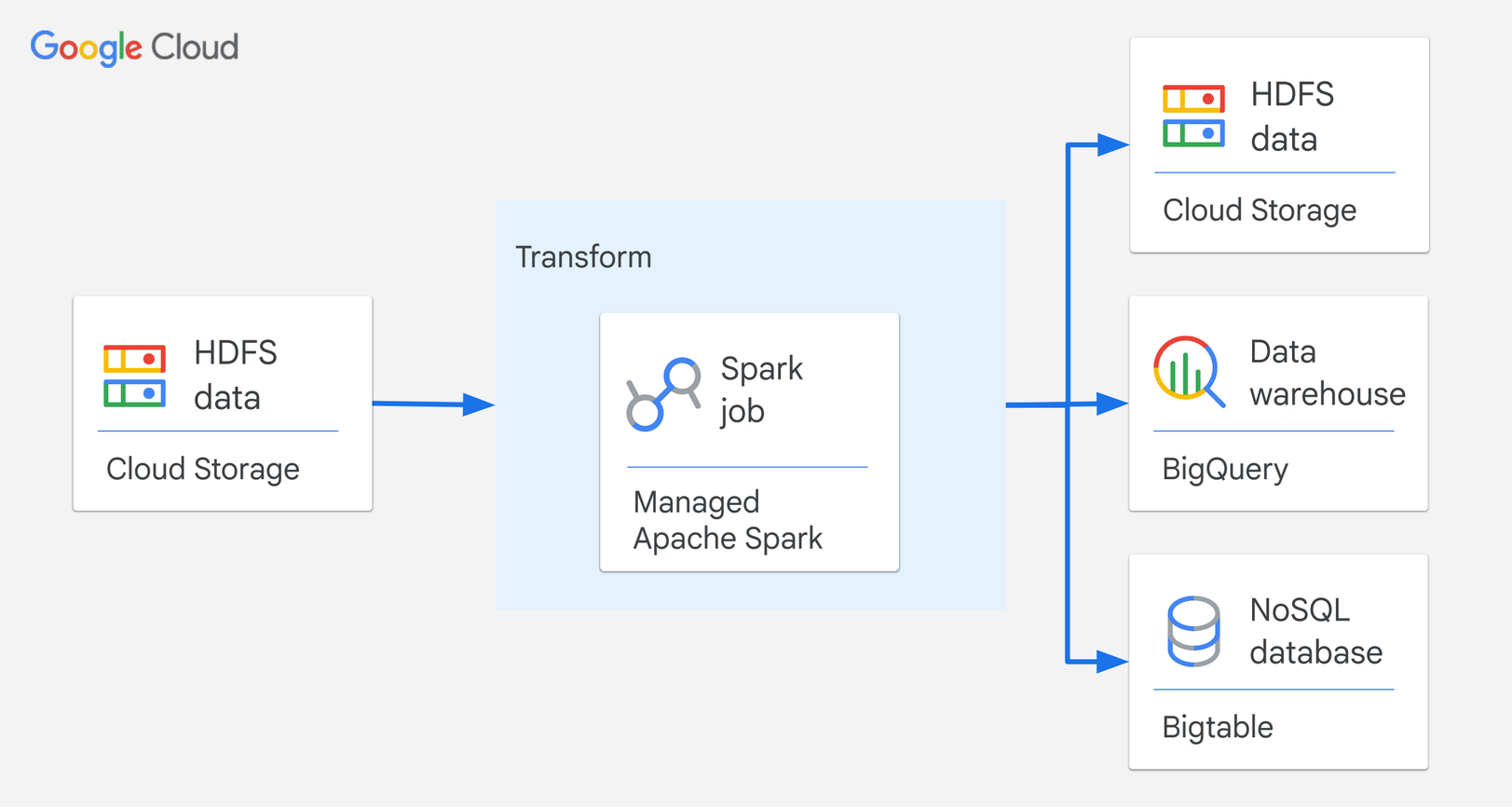
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Run your Apache Hadoop and Spark workloads on Managed Service for Apache Spark



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Managed Service for Apache Spark is the managed service for data processing with Hadoop and Spark

Runtimes for Compute Engine, GKE, Serverless for Spark

Workflow templates with dependencies

Integration with other storage services for HDFS replacement



Managed Apache Spark

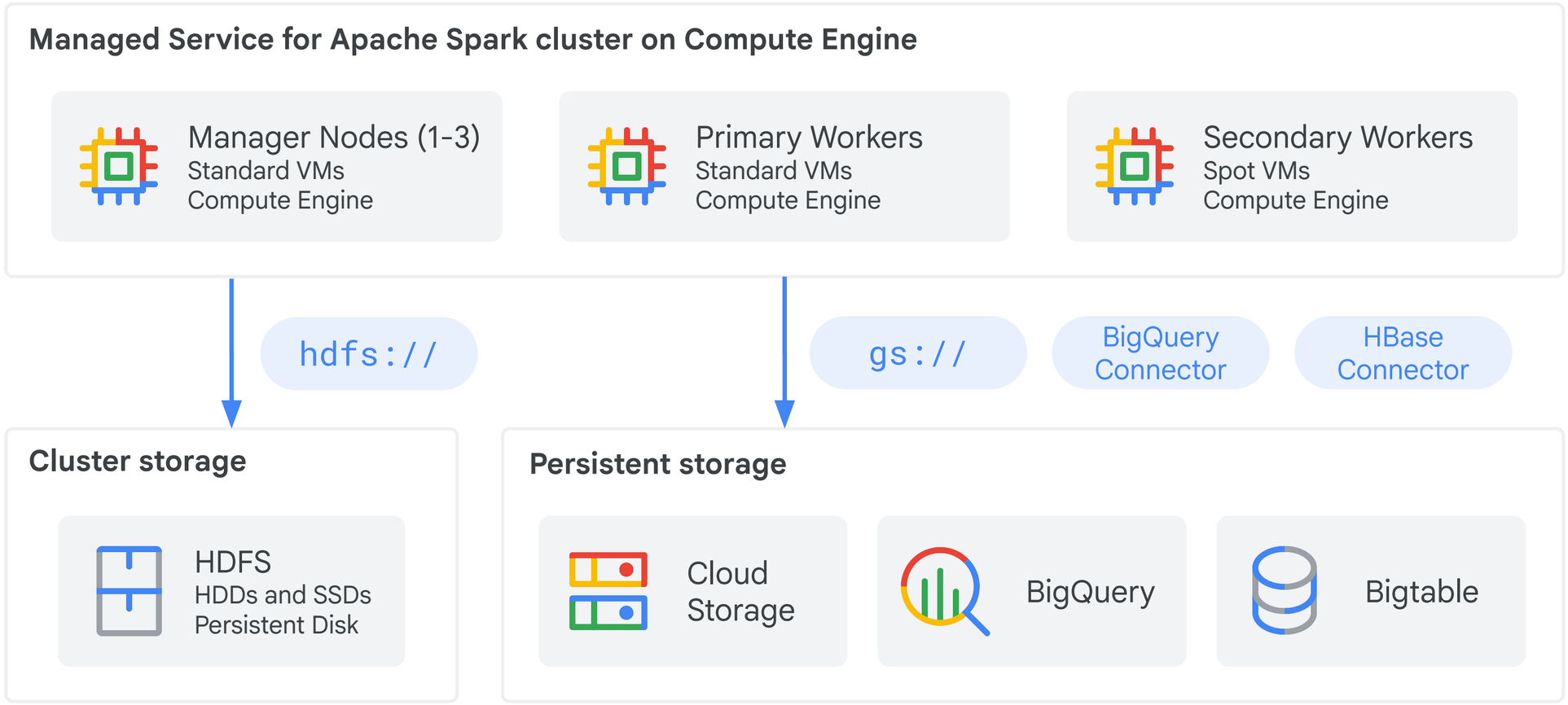
Rich open-source ecosystem (Hadoop, Spark, Pig, Hive pre-installed)

Reactive **autoscaling** policies

Permanent and **ephemeral** clusters with scheduled deletion

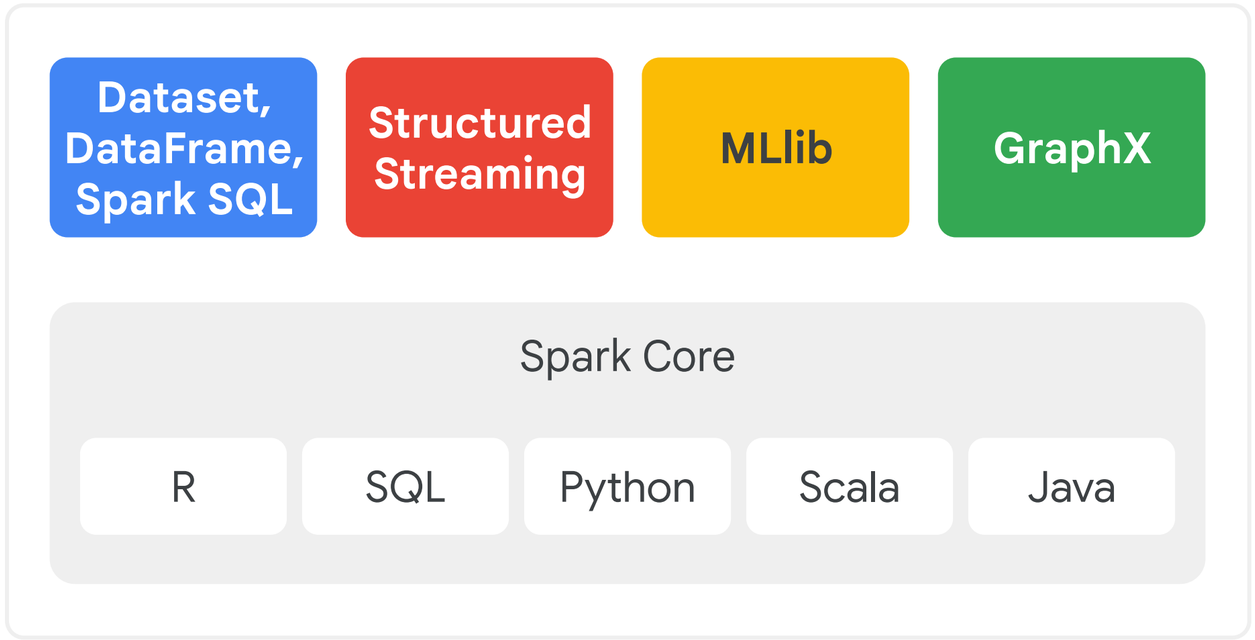
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Managed Service for Apache Spark clusters can use HDFS on persistent disks or other Google Cloud storage services



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Apache Spark is a popular framework for many data processing tasks



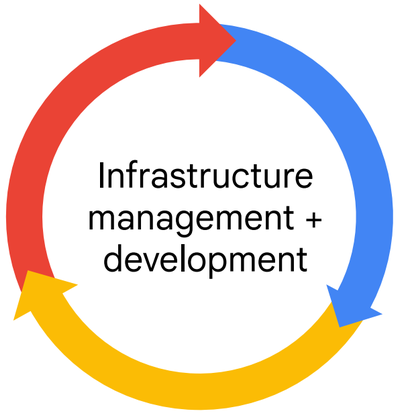
- Data engineering
- ML model training
- Batch inferencing
- Graph processing
- Geospatial analytics
- Time series analytics

... and more

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Run Spark workloads without cluster management using Managed Service for Apache Spark serverless

Before



- ✗ Manage Clusters
- ✓ Write code
- ✗ Decide infrastructure
- ✗ Pay while it's running
- ✗ Wait for your job to run (queued up)

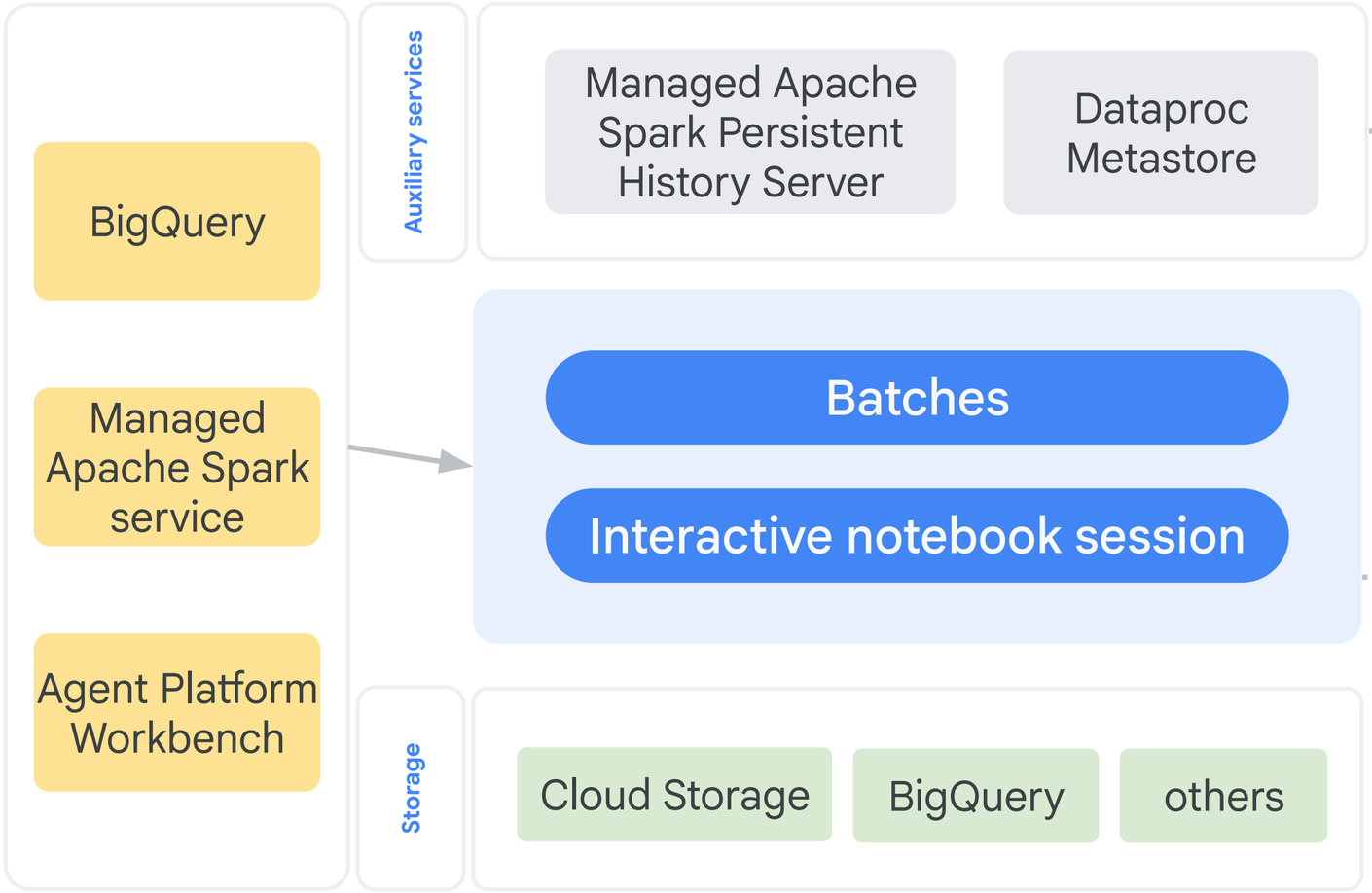
Using Managed Apache Spark serverless



- ✓ Auto-scaling
- ✓ No infrastructure to tune
- ✓ No clusters to manage
- ✓ Only pay for the execution time (no startup or shutdown cost)
- ✓ No resource contention
- ✓ Speed to production
- ✓ Batches, interactive notebooks, Agent Platform Pipelines

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Managed Service for Apache Spark serverless provides rich integrations with Google Cloud services



Managed Service for Apache Spark Persistent History Server:

- Persistent store for logs and information about completed Spark jobs.

Dataproc Metastore:

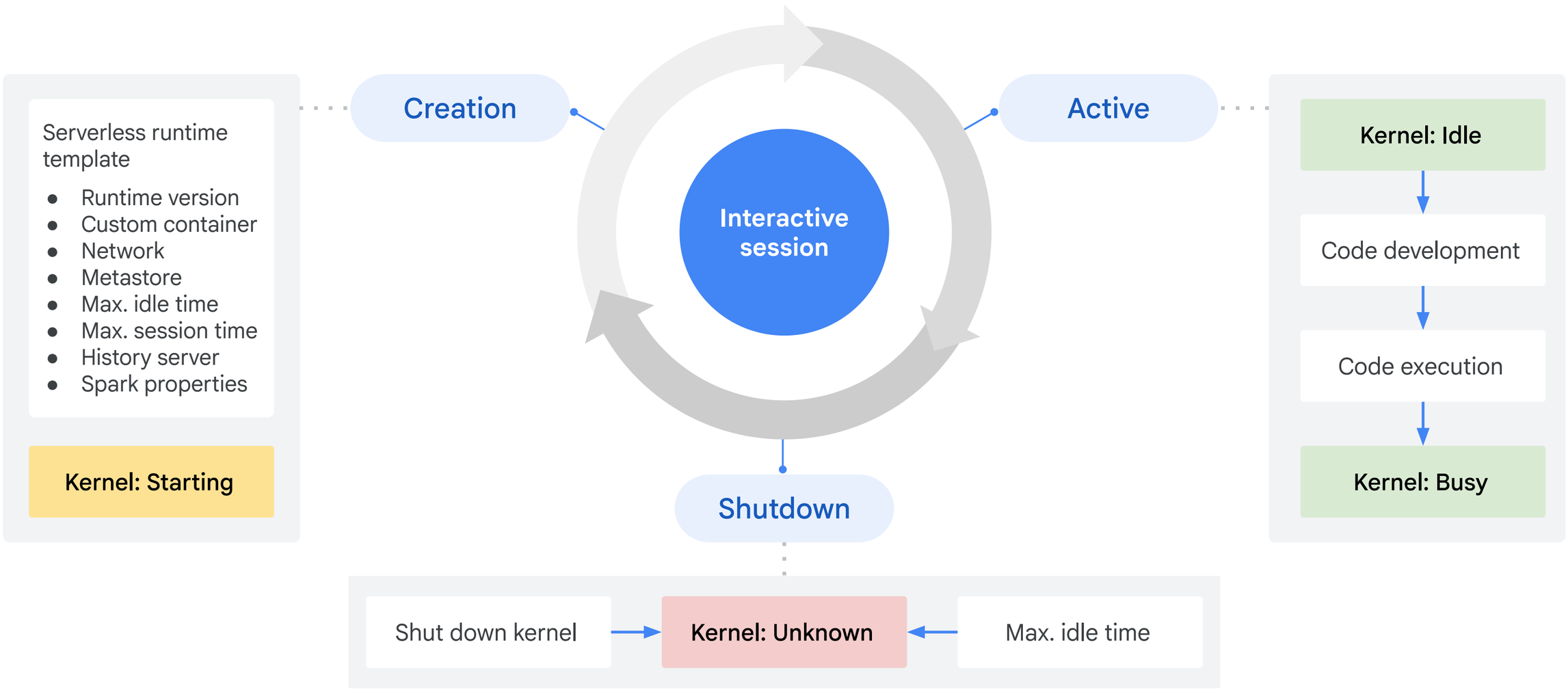
- Metadata for Hive tables.

Under the hood:

- Creates ephemeral cluster using Managed Service for Apache Spark custom images.
- Submits job for execution.
- Deletes cluster after execution.

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The lifecycle of an interactive notebook session



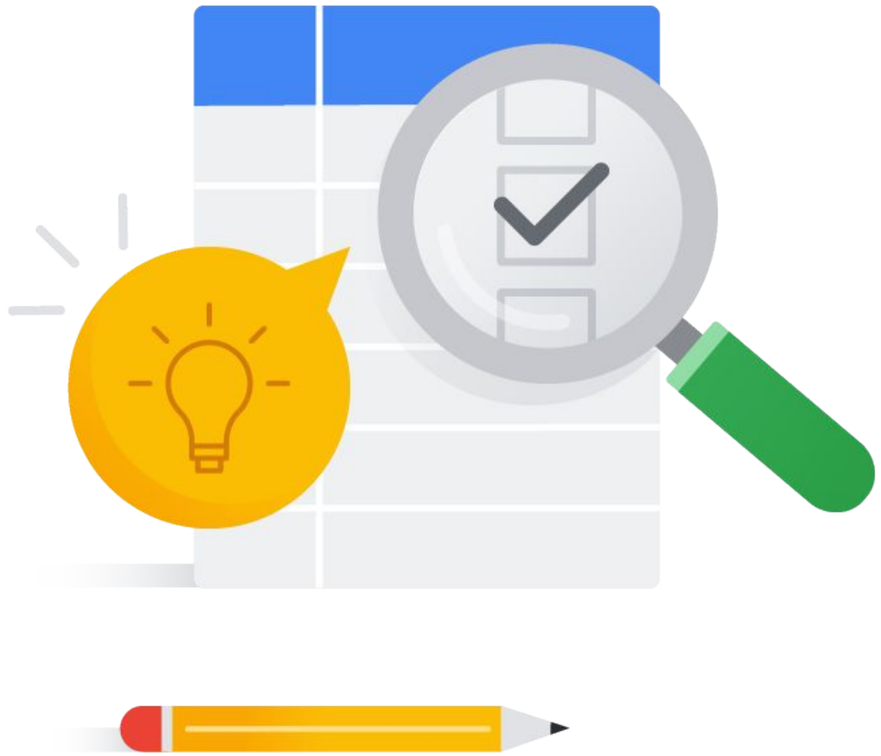
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Lab (optional): Use Serverless for Apache Spark to Load BigQuery

30 min

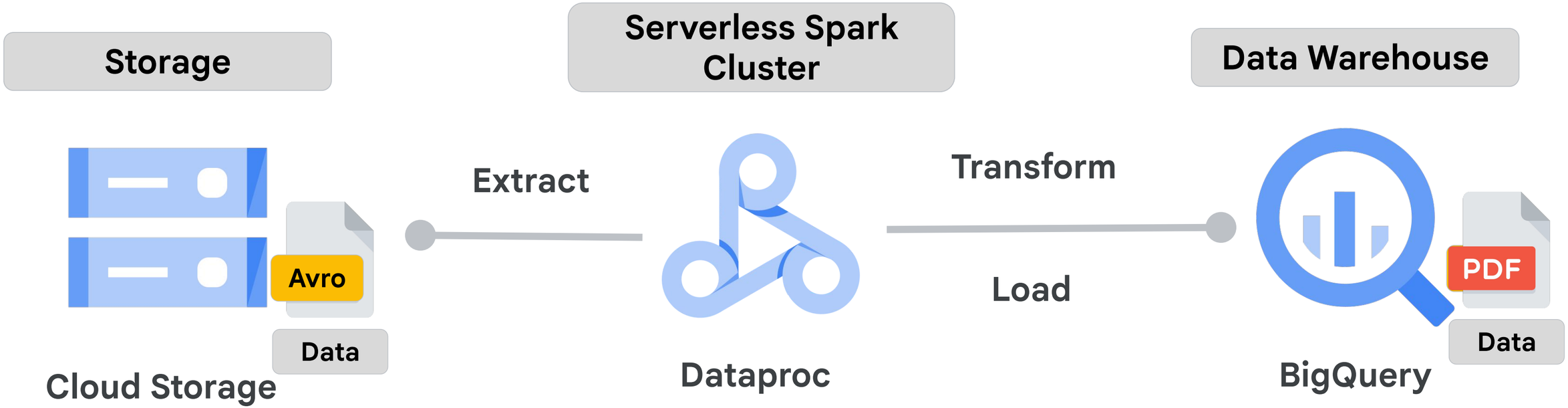
Learning objectives

- Configure the environment.
- Download lab assets.
- Configure and execute the Spark code.
- View data in BigQuery.



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Lab (optional): Use Serverless for Apache Spark to Load BigQuery



Google Cloud

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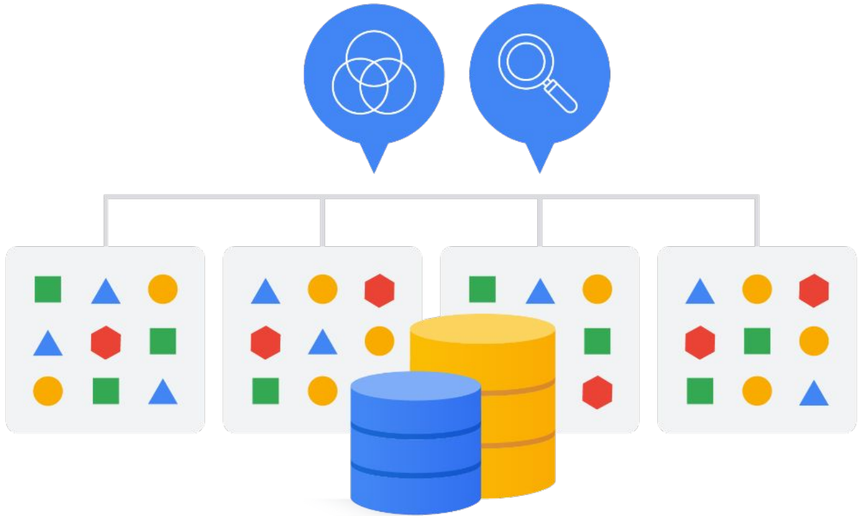
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Batch processing versus streaming data processing

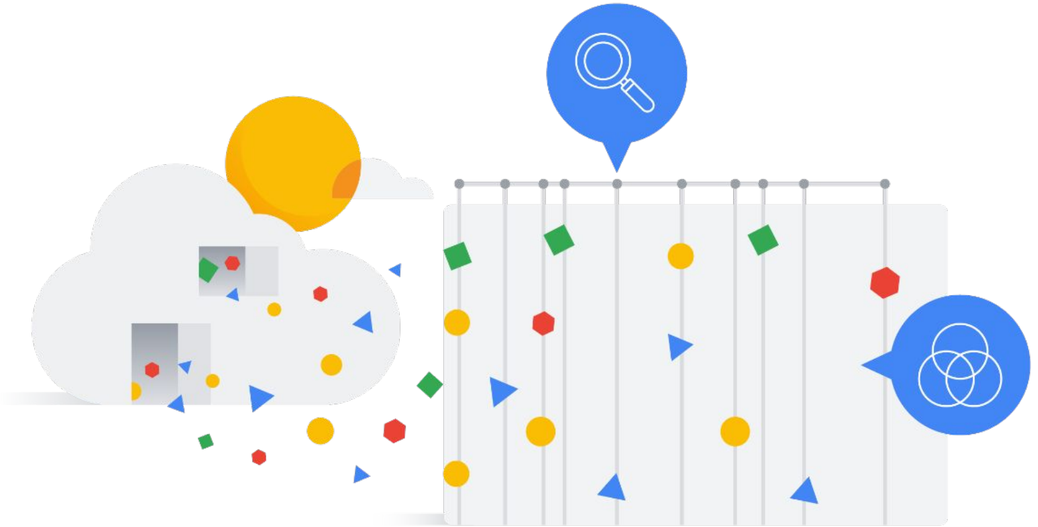


Batch processing

Processing and analysis happens on a set of stored data.

Payroll system

Billing system



Streaming data processing

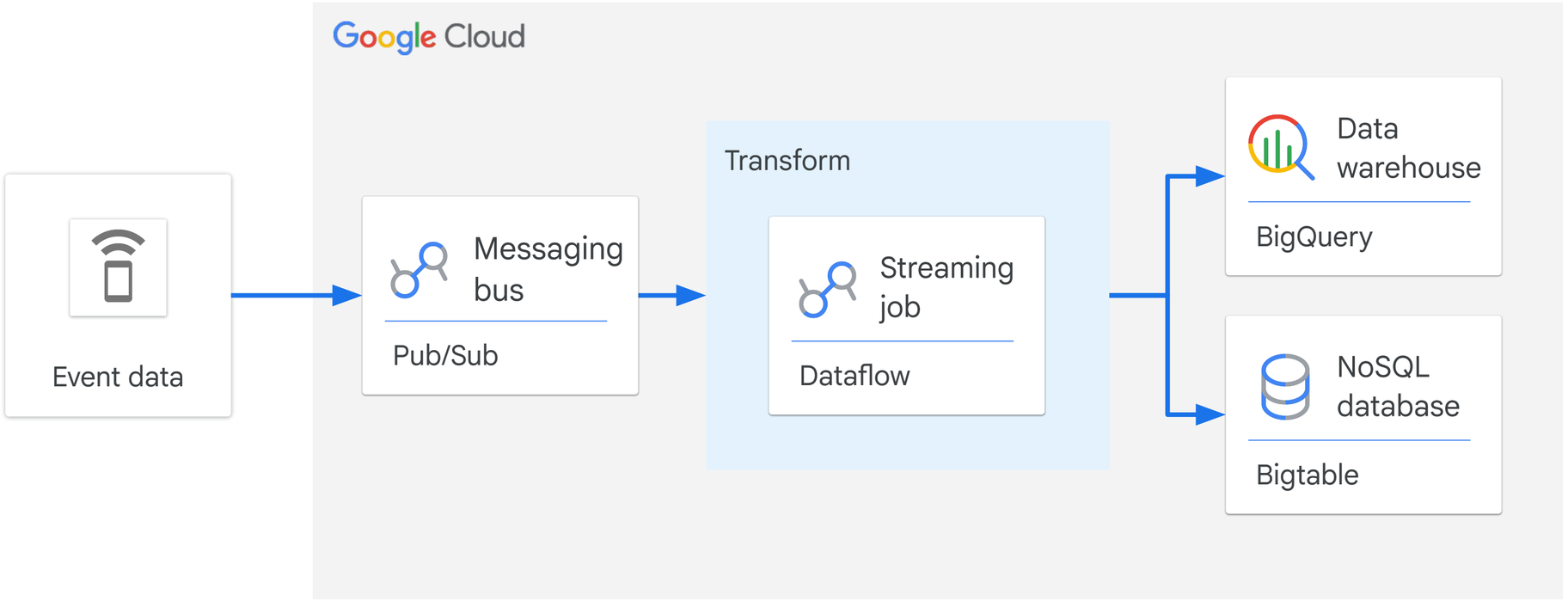
A flow of data records generated by various data sources.

Fraud detection

Intrusion detection

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Streaming ETL workloads require continuous data ingestion, processing, and near real-time analytics



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Example: Streaming and transforming messages from Pub/Sub to BigQuery

```
with beam.Pipeline(options=pipeline_options) as pipeline:

   # Read messages from Pub/Sub
   messages = pipeline | 'read' >> ReadFromPubSub(topic='<PUBSUB_TOPIC>')

   # Parse messages
   def parse_message(message):
       return json.loads(message.decode('utf-8'))

   parsed_messages = messages | 'parse' >> beam.Map(parse_message)

   # Write parsed messages to BigQuery
   parsed_messages | 'write' >> WriteToBigQuery(
       table='<BIGQUERY_TABLE>',
       schema=table_schema,
       create_disposition=BigQueryDisposition.CREATE_IF_NEEDED,
       write_disposition=BigQueryDisposition.WRITE_APPEND
   )

# Run the pipeline
pipeline.run()
```

`ReadFromPubSub()`:

- Retrieves messages from a Pub/Sub topic or subscription.

`beam.Map()`:

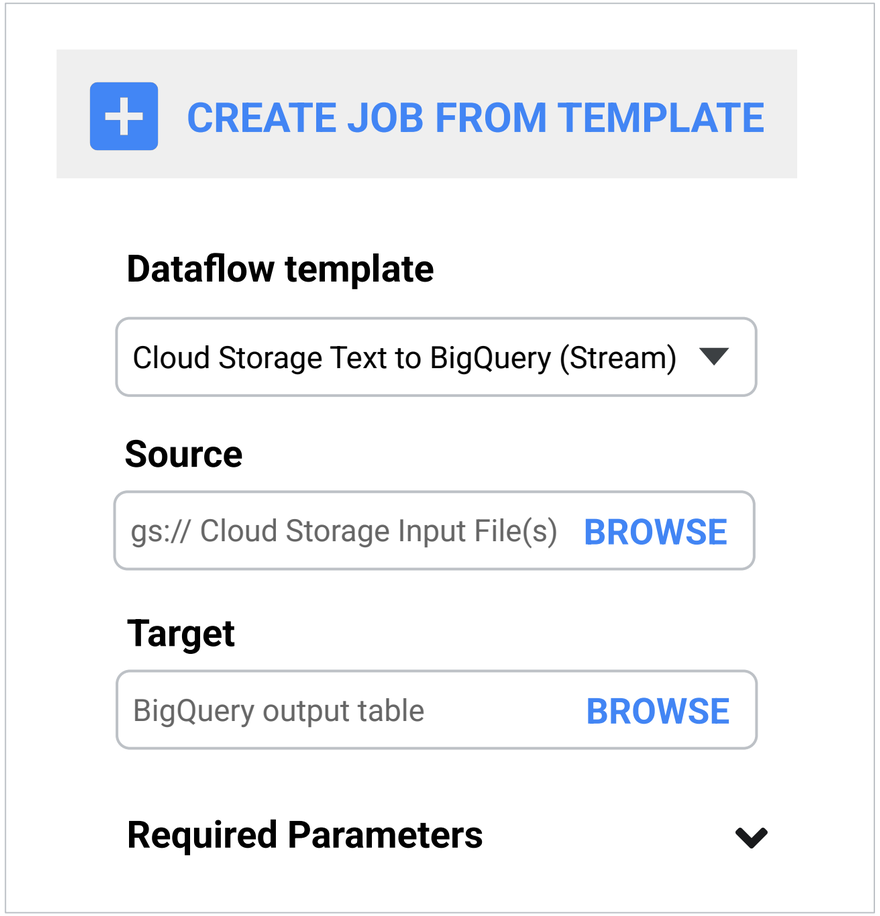
- Applies a specified transformation to the message.

`WriteToBigQuery()`:

- Writes the transformed message into a BigQuery table.

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Use Dataflow templates for recurring pipeline executions with different parameters



Benefits of using templates

- Pipeline design is separate from deployment.
- Parameters customize the pipeline and make it reusable with different inputs.
- Templated pipelines can be deployed from the Google Cloud console, command-line interface, or API.
- Use Google-provided pre-built templates for common scenarios.

You can create your own custom templates.

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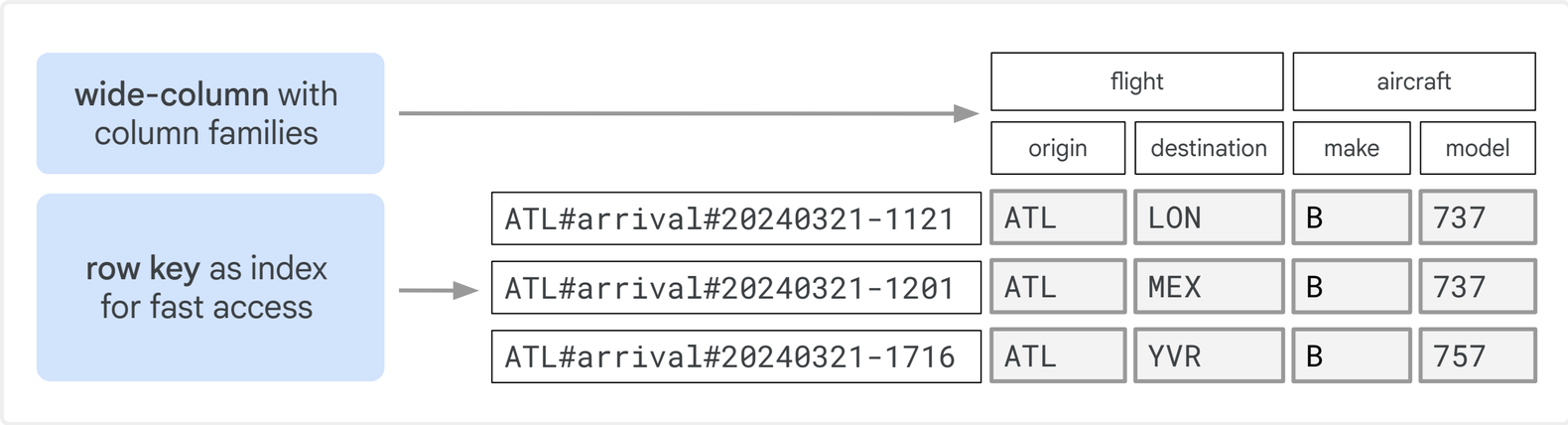
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Use Bigtable as a sink in your streaming data pipeline for millisecond latency analytics




```
// first row output
ATL#arrival#20240321-1121  column=flight:origin, timestamp: 1711020073, value=ATL
ATL#arrival#20240321-1121  column=flight:destination, timestamp: 1711020073, value=LON
ATL#arrival#20240321-1121  column=aircraft:make, timestamp: 1711020073, value=B
ATL#arrival#20240321-1121  column=aircraft:model, timestamp: 1711020073, value=737
```

- NoSQL queries over large datasets (petabytes)
- High throughput with millisecond latency
- Uses open-source HBase API for integration

Applications:

- Time-series data
- IoT
- Financial data
- Machine learning

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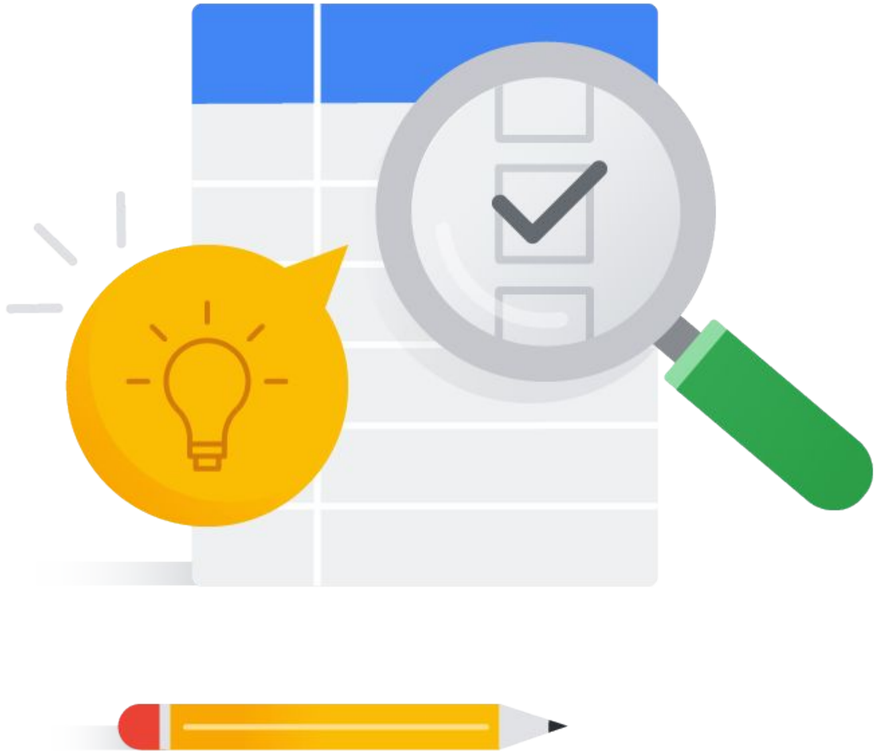
Compare ETL processing options

	Dataprep	Cloud Data Fusion	Managed Service for Apache Spark	Dataflow
Applications	Data wrangling	Data integration	ETL workloads	ETL workloads
Integrations	Cloud Storage, BigQuery, others	hybrid, multicloud environments	Cloud Storage, BigQuery, other OSS	Pub/Sub, Cloud Storage, BigQuery, other OSS
Open source	–	CDAP	Apache Hadoop, Spark, other OSS	Apache Beam
Velocity	Batch	Batch, stream	Batch, stream	Batch, stream (recommended)
Serverless	yes	no	no (except Serverless Spark)	yes

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Lab: Creating a Streaming Data Pipeline for a Real-Time Dashboard with Dataflow

45 min

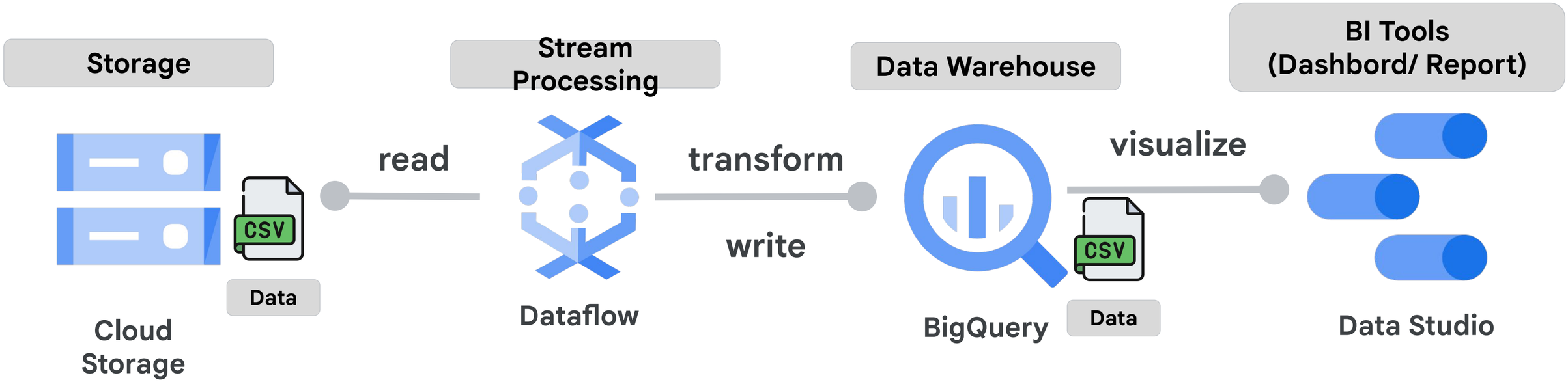


Learning objectives

- Create a Dataflow job from a template.
- Stream data via Dataflow pipeline into BigQuery.
- Monitor a Dataflow pipeline in BigQuery.
- Analyze results with SQL.
- Visualize key metrics in Data Studio.

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Lab: Creating a Streaming Data Pipeline for a Real-Time Dashboard with Dataflow




Google Cloud